GlucoSense GS-1

A single-chip wearable glucose sensor interface with Bluetooth Low Energy and an on-chip model of the pancreatic islet

Research and development proposal with chip layout information

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Submitted to	The Project Director, for initial review and funding approval
Duration	30 months
Requested, fabrication and prototyping	45,828 EUR, plus staff at host scales (Section 11)
Technology readiness	TRL 3 today; TRL 5 at the end of the project
Design repository	https://github.com/asifnazquaidian/glucosense-gs1
Live application	https://glucosensor.netlify.app
Earlier chip by the applicant	https://github.com/asifnazquaidian/neurobit64

Executive summary

The problem. 589 million adults live with diabetes and 252 million of them do not know it [1]. Continuous glucose sensors worn on the skin are the most effective tool for managing the disease, but the electronics inside them are proprietary single chips that report one number, glucose. Researchers who want to test a new sensor or a new algorithm cannot obtain such a chip and build bench prototypes from development boards instead.

The proposal. GS-1 is one open, documented chip that makes a complete wearable: it biases an electrochemical glucose sensor, measures its nanoampere current, temperature and battery, and sends the result over Bluetooth Low Energy to the GlucoSense application. Sensor, chip, antenna, crystal and a coin cell are the whole patch.

The innovation. GS-1 carries a hardware model of the pancreatic islet. The Hybrid Excitable Cell (HEC) model of pancreatic α, β and δ cells from the applicant's PhD thesis and publications [4, 5] runs on the chip and is driven by the measured glucose. With every reading the chip reports how a model islet responds to the wearer's glucose: a physiological reference for insulin and glucagon demand next to the raw number.

Evidence already in hand (public, reproducible; Section 4):

- the GlucoSense application, live at https://glucosensor.netlify.app, with signal conditioning, calibration, alerts, reports, an islet-model view and both Bluetooth modes;
- the complete digital part of the chip in Verilog (7,900 logic cells), identical bit for bit to two independent software models, including 9,600 Bluetooth on-air bits, and read correctly by the application;
- the islet block within one spike per period of the thesis model from 40 to 400 mg/dL;
- the full-chip floorplan: 1.6 mm $\times$ 1.6 mm die, ten blocks, 28 pads;
- a clean, manufacturable layout of the applicant's previous chip from the same design flow.

The plan. Three silicon steps with two go/no-go decisions: the digital core on a low-cost shared wafer with a working wearable demonstrator; the analog front end as a test chip; the integrated GS-1 with radio. All results, design files and software are released openly.

What is not claimed. No silicon exists yet; the analog and radio circuits are specified and floorplanned, not designed; nothing has been tested with a real sensor or on people. GlucoSense is a research platform, not a medical device. Section 12 lists completed and funded work line by line.

Contents

1 Problem and significance **2**

2 State of the art and the gap **3**

3 Objectives and success criteria **3**

4 Preliminary work and results **3**

4.1 The GlucoSense application and its live test . . . 3

4.2 Verification of the chip's digital core . . . 4

4.3 Track record: the same flow has produced a clean layout . . . 4

4.4 Technology readiness . . . 4

5 Technical description **4**

5.1 System and chip architecture . . . 4

5.2 The HEC pancreatic islet model on the chip . . . 5

5.3 Chip layout information . . . 7

6 Methodology and work plan **9**

6.1 Work packages . . . 9

6.2 Schedule and milestones . . . 9

6.3 Methods . . . 10

7 Impact **10**

7.1 Dissemination, exploitation and intellectual property . . . 10

8 Data management and open science **10**

9 Ethics, safety and regulatory position **10**

10 Risk management **11**

11 Resources and budget **11**

11.1 Team . . . 11

11.2 Fabrication and prototyping budget . . . 11

12 Status declaration: completed and funded work **12**

1 Problem and significance

Diabetes affects 589 million adults (one in nine) and is projected to reach 853 million by 2050 [1]. Good control depends on knowing glucose continuously: international consensus targets are now expressed as time in the 70–180 mg/dL range, measured by continuous sensors [2], and summarised by indicators such as the glucose management indicator [3].

A continuous sensor patch contains an electrochemical sensor, a potentiostat that holds the sensor at its working potential and measures a current of a few nanoamperes, a converter, a radio and a battery. To reach the size, two-week battery life and unit cost of a disposable patch, all of the electronics must be a single custom chip. Those chips exist only inside commercial products, with closed protocols.

Two consequences motivate this project. First, academic and clinical researchers have no open chip-level platform on which to evaluate new sensor chemistries, calibration methods or algorithms; they use multi-chip boards built from a

general-purpose electrochemical front end, a microcontroller and a radio module, which cannot be worn for long and do not represent a product. Second, today's devices report glucose alone. They say nothing about how the body's own regulation is responding, which is the information that distinguishes early impaired regulation from normal variation.

2 State of the art and the gap

Approach	Strength	Limitation addressed by GS-1
Commercial continuous glucose monitors	Clinically validated, single-chip, two-week wear	Closed chip and protocol; not available as a research platform; glucose only
General-purpose electrochemical front-end chips with a separate microcontroller and Bluetooth module	Available, flexible	Three or more chips, tens of square centimetres of board, milliampere-level power; not a wearable patch
Published research sensor-interface chips	Low noise and power	Rarely with an integrated radio and application; design files not released
Glucose–insulin simulation models on a computer [9]	Physiological insight	Run off-line on recorded data, not on the sensor in real time

The gap: no open sensor-interface chip combines the measurement chain, the radio link, an open application and a real-time physiological model. GS-1 is designed to fill it. The HEC model is well suited to hardware: it is a two-variable model in the family of [8] that the applicant extended to cardiac and pancreatic cells [4, 5, 6, 7] and that needs only additions, shifts and one multiplication per step.

3 Objectives and success criteria

	Objective	Measurable success criterion	Month
O1	Fabricate the digital core and demonstrate a wearable-size prototype with it	Chip output identical to the reference model; readings shown live in the application in connected and beacon mode	12
O2	Design and measure the analog front end	Input-referred noise below 10 pA rms; current error within 1 % of reading or 20 pA over $\pm$180 nA	18
O3	Integrate the single-chip GS-1 with radio	Bluetooth packets received by an unmodified phone at 2 m with at least 99 % delivery; average supply current at most 10 µA at one reading per minute	30
O4	Validate the HEC islet model against published glucose–insulin data	Model insulin response reproduces reference responses to meal tests within an error bound fixed in advance; method and data published	18
O5	Show in-vitro measurement accuracy	Glucose solutions from 40 to 400 mg/dL within the limits of ISO 15197 [11] against a laboratory analyser	30
O6	Release an open platform	Design files, models, application and measurement data public under open licences; two journal papers	30

4 Preliminary work and results

All items in this section exist today and can be reproduced from the design repository (`https://github.com/asifnazquaidian/glucosense-gs1`; folders are named in brackets).

4.1 The GlucoSense application and its live test

The application [13] is an installable web application for phone and PC. The reviewer can run the live test now: open `https://glucosensor.netlify.app`, press *Start simulator*, and the complete chain runs in the browser.

- *Live, Signal, Log, Reports:* packet handling with loss detection, despiking and smoothing, temperature compensation, finger-stick calibration, trend and alerts, time-in-range statistics [2, 3] and CSV export.
- *Islet:* the HEC islet model for every reading. It displays the chip's three islet bytes when they are present and otherwise runs the identical arithmetic in the application.
- *Device:* three ways to receive data: a Bluetooth connection (GATT notification); beacon mode, which listens for the chip's advertising packets without connecting; and a demonstration connection, which lists nearby Bluetooth devices, shows the chosen device's name and signal strength, and runs a clearly labelled random signal for training and illustration.

The simulator and the chip produce the same packet (Table 1), so every stage after the radio is already exercised.

Table 1: The GlucoSense measurement packet (little-endian). GS-1 produces exactly this packet and appends three islet bytes.

Byte	Field	Type	Meaning
0–1	seq	uint16	packet counter (gaps show lost packets)
2–5	uptime	uint32	device clock, ms
6–9	current	int32	sensor current, pA
10–11	temp	int16	temperature, 0.01 °C
12	battery	uint8	percent
13	flags	uint8	bit 0 warm-up, bit 1 ADC saturated, bit 2 low battery
14–16	islet	3 × uint8	*new in GS-1:* α, β, δ cell spike counts

4.2 Verification of the chip's digital core

Test	Result
Verilog chip against the JavaScript reference model: glucose swing, sensor overload in both directions, flat battery, register writes, out-of-range interval command [`design/results`]	10 of 10 frames and 30 of 30 advertising packets identical (9,600 on-air bits)
The same in self-test mode (on-chip test modulator)	identical
Verilog chip against a separately written Python model [`test`]	10 of 10 frames identical
Chip packets through the application's decoder [`app/proto.js`]	10 of 10 decoded and re-encoded identically
Chip on-air bits through a separately written receiver model (de-whitening, CRC check, parsing) into the running application's beacon mode	30 of 30 packets pass the CRC; 10 of 10 readings stored with identical current, temperature and islet bytes; repeats on the other two channels and a second sensor ignored
Application islet arithmetic against the chip model	identical over 400 periods
HEC islet integer arithmetic against the floating-point thesis model, 40–400 mg/dL	mean difference 0.12 spikes per period, largest 1
Synthesis, SkyWater 130 nm high-density library	7,900 cells, 790 flip-flops, 64,278 µm^2 cell area

The Bluetooth packet follows the Bluetooth Core Specification [12] (preamble, access address `0x8E89BED6`, CRC-24, whitening by channel). Three independently written pieces of software agree on it. Reception over the air by a phone has not been tested, because that needs the radio; in the application test the browser's Bluetooth scanning interface was replaced by a stand-in that delivers the simulated packets.

4.3 Track record: the same flow has produced a clean layout

The applicant's NeuroBit64-S chip (64 spiking neurons, 3,172 cells) was taken from Verilog to a finished layout in SkyWater 130 nm with the open-source Tiny Tapeout flow [15]: 0 design-rule violations, 0 layout-versus-schematic errors, timing met at 50 MHz with 4.45 ns margin in the worst corner. Design files and layout reports are public [16]. A second, larger chip (64 HEC neurons, 14,072 cells) followed and is also free of design-rule and layout-versus-schematic errors. The GS-1 digital core is packaged for the same flow (4×2 tiles, about 42 % utilisation); starting that run is the first task of the project.

4.4 Technology readiness

Element	Evidence	TRL today
Application software	Running publicly; tested end to end with simulated sensors	4
Digital core and islet block	Verified in simulation against three models; synthesised	3
Analog front end, radio, power management	Specified, area-budgeted, floorplanned	2
Complete chip	Target of this project: integrated prototype tested in the laboratory	5 (month 30)

5 Technical description

5.1 System and chip architecture

Once per sample interval (100 ms to 60 s, set from the application) the chip does the following; the figures are for the 8 MHz clock of the present design.

1. **Convert** three channels with one incremental sigma-delta converter [10]: sensor current, temperature, battery. Each

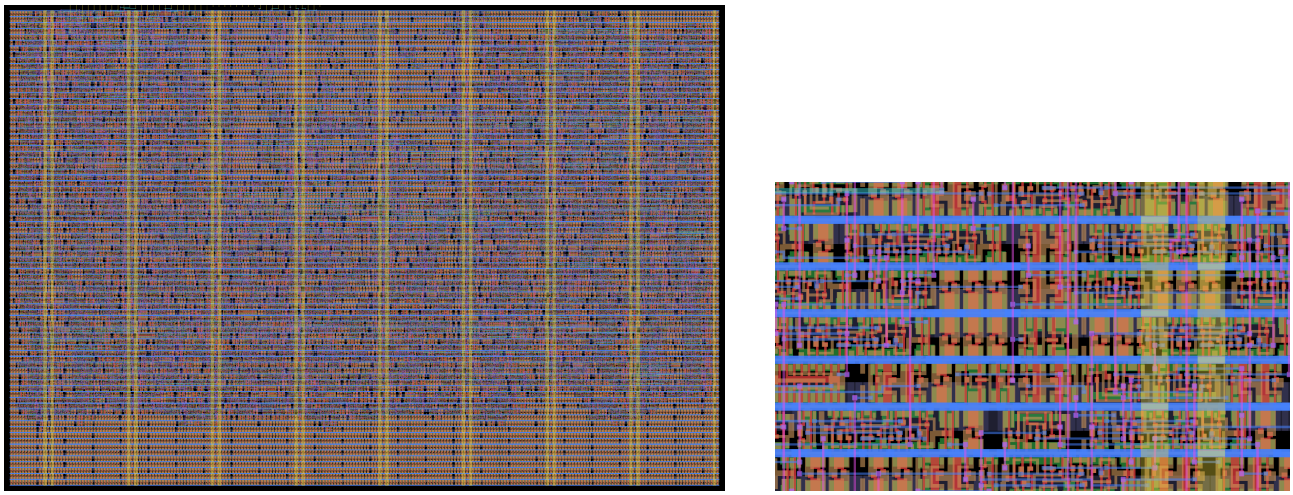


Figure 1: Finished layout of the applicant's previous chip, NeuroBit64-S [16], made with the flow proposed for the GS-1 digital core (full block and close-up). This is *not* the GS-1 layout.

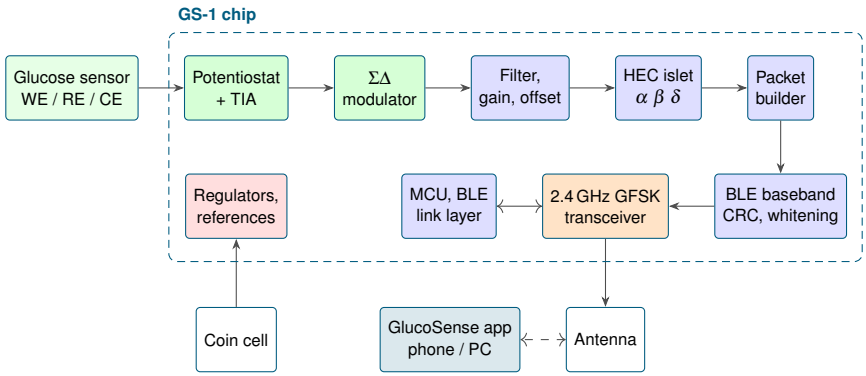


Figure 2: Signal path. Green: analog. Blue: digital. Orange: radio. Temperature and battery share the converter through an input multiplexer.

conversion is 1024 modulator clocks at 250 kHz (4.1 ms). The analog modulator supplies one bit per clock; the digital core supplies its clock and reset and filters the bits with two cascaded integrators.

2. **Scale** each result with a per-chip gain and offset to pA, 0.01 °C and percent.
3. **Run the islet model** for one period of each cell type (Section 5.2).
4. **Send**: (a) a Bluetooth LE advertising packet on channels 37, 38 and 39; (b) the same data on a UART for test and for a companion radio in the first two project stages.

The whole cycle is 16 ms of activity; for the rest of the interval only the sensor bias and a timer are on, which is where the long battery life comes from.

Two radio modes. *Beacon mode* sends each reading in an advertising packet. It needs a transmitter only and no protocol software; it is implemented in the verified digital core and in the application. *Connected mode* (a GATT notification) needs a receiver and the Bluetooth link layer, which run on the processor block of the floorplan; until that exists, a commercial Bluetooth module fed by the chip's UART provides it.

5.2 The HEC pancreatic islet model on the chip

The Hybrid Excitable Cell model was proposed in the applicant's PhD thesis [4] as one compact model for neurons, cardiac myocytes and pancreatic cells, and published in [5, 6, 7]. For pancreatic cells it reads

$$C V' = k f (V - V_r)(V - V_t) - u + V_\alpha + V_\beta + R I, \qquad u' = a\big(b(V - V_r) - u\big),$$

$$V_\alpha = V_{glu}\, e^{-(t - n\tau)/T_k}, \qquad V_\beta = V_{ins}\, e^{-(t - n\tau)^2/T_{Ca}}, \qquad V_{th} = V_t + V_\alpha + V_\beta,$$

with a spike and reset ($V \leftarrow c$, $u \leftarrow u + d$) when $V \geq V_{th}$. The glucagon- and insulin-induced potentials V_α, V_β raise the threshold and shape the bursts. The chip uses the thesis parameters for the three cell types:

Cell	a	b	d	k	f	V_{glu} (mV)	V_{ins} (mV)
α	0.03	−2	100	1	0.1	0	0
β	0.02	−5	50	3	0.05	35	30
δ	0.02	−1	60	0.3	0.2	35	40

$T_k = T_{Ca} = 250$ steps follow from the thesis values of 5 MΩ and 50 µF. Values the table does not list ($C = 100$, $V_r = -60$ mV, $V_t = -40$ mV, $c = -50$ mV) are taken from the thesis HEC code; the period τ is 1024 steps.

Extension made in this proposal. In the thesis the excitation RI is a given input. Here it is tied to the measured glucose: β and δ cells are driven in proportion to glucose above a programmable threshold (default 80 mg/dL), α cells in proportion to glucose below a second threshold (default 90 mg/dL). This coupling is a design hypothesis; testing and tuning it against published data is Objective O4.

Hardware. The three cells share one arithmetic circuit and take 1024 steps each per reading (0.8 ms). All quantities are integers in units of 1/64 mV, and the two exponentials are computed with shifts and adds only. Figure 3 compares the chip arithmetic with the floating-point thesis model.

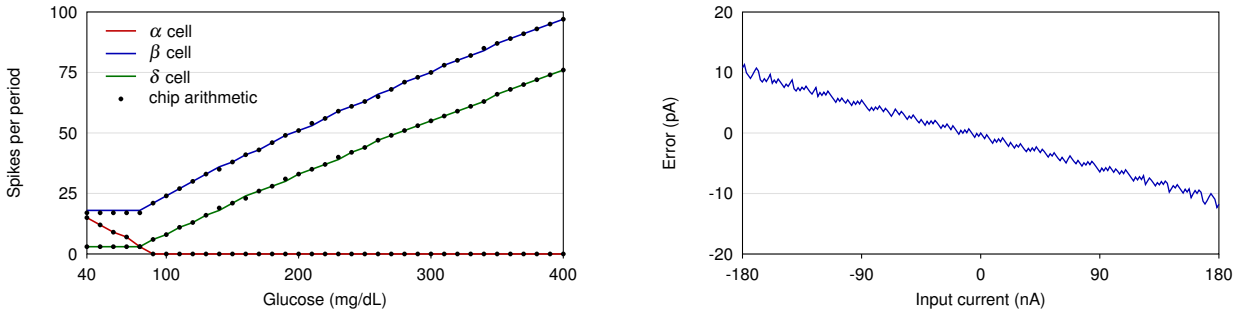


Figure 3: Left: islet response against glucose; lines are the thesis model in floating point, dots the integer arithmetic of the chip. Right: error of the current measurement (ideal modulator, digital filter and scaling as on the chip) after gain trim.

What the islet output is for. Figure 4 shows the model in a closed loop as a virtual patient: islet activity sets insulin and glucagon action, which change glucose after two meals. With the normal β-cell response glucose stays between 95 and 164 mg/dL; with the β-cell drive reduced to 35 % it reaches 215 mg/dL and returns slowly. On the chip, the same model run on the wearer's measured glucose gives a reference response. A persistent gap between that reference and what the glucose actually does is the signal of interest for early detection of impaired regulation, in line with the anomaly-detection work in [5]. The loop constants of Figure 4 are illustrative and are not fitted to patient data.

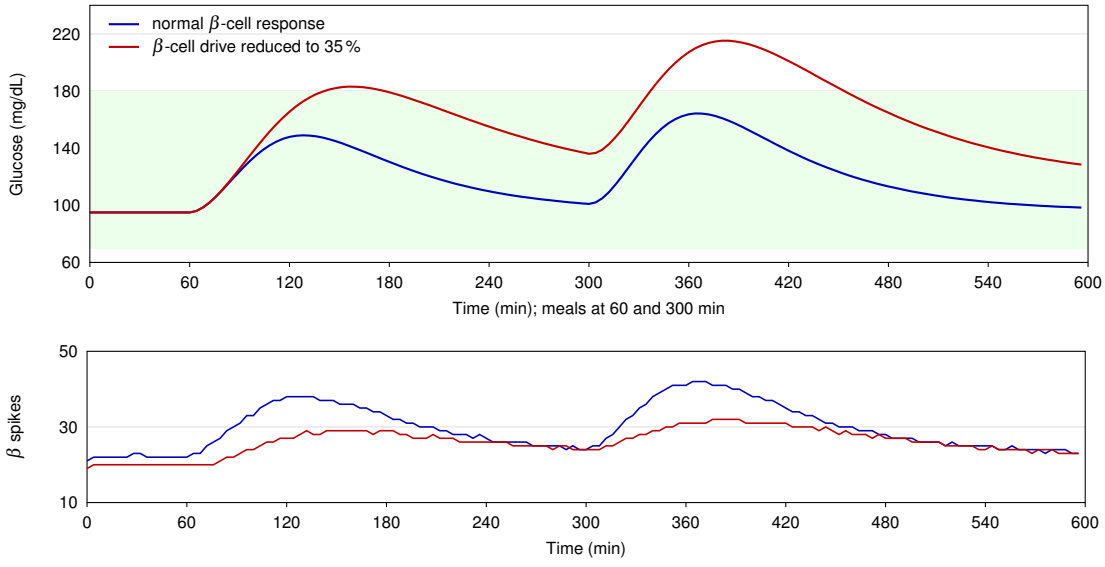


Figure 4: Virtual patient built on the HEC islet model (chip arithmetic). Top: glucose; the band is the 70–180 mg/dL target range. Bottom: β-cell spikes per period.

5.3 Chip layout information

Technology	TSMC 65 nm LP mixed-signal/RF CMOS (shared-wafer access through Europractice [14]); 2.5 V I/O transistors for the sensor interface
Die	1600 µm × 1600 µm = 2.56 mm^2; core area inside the pad ring 1340 µm × 1340 µm
Pads	28, opening 70 µm × 90 µm, pitch 170 µm, seven per side
Package	QFN-28, 4 mm × 4 mm for prototypes; bare die on flexible board for the patch
Supply	1.8 to 3.6 V battery; internal regulators for analog, digital and RF
External parts	32 MHz crystal, antenna and matching, five decoupling capacitors, sensor

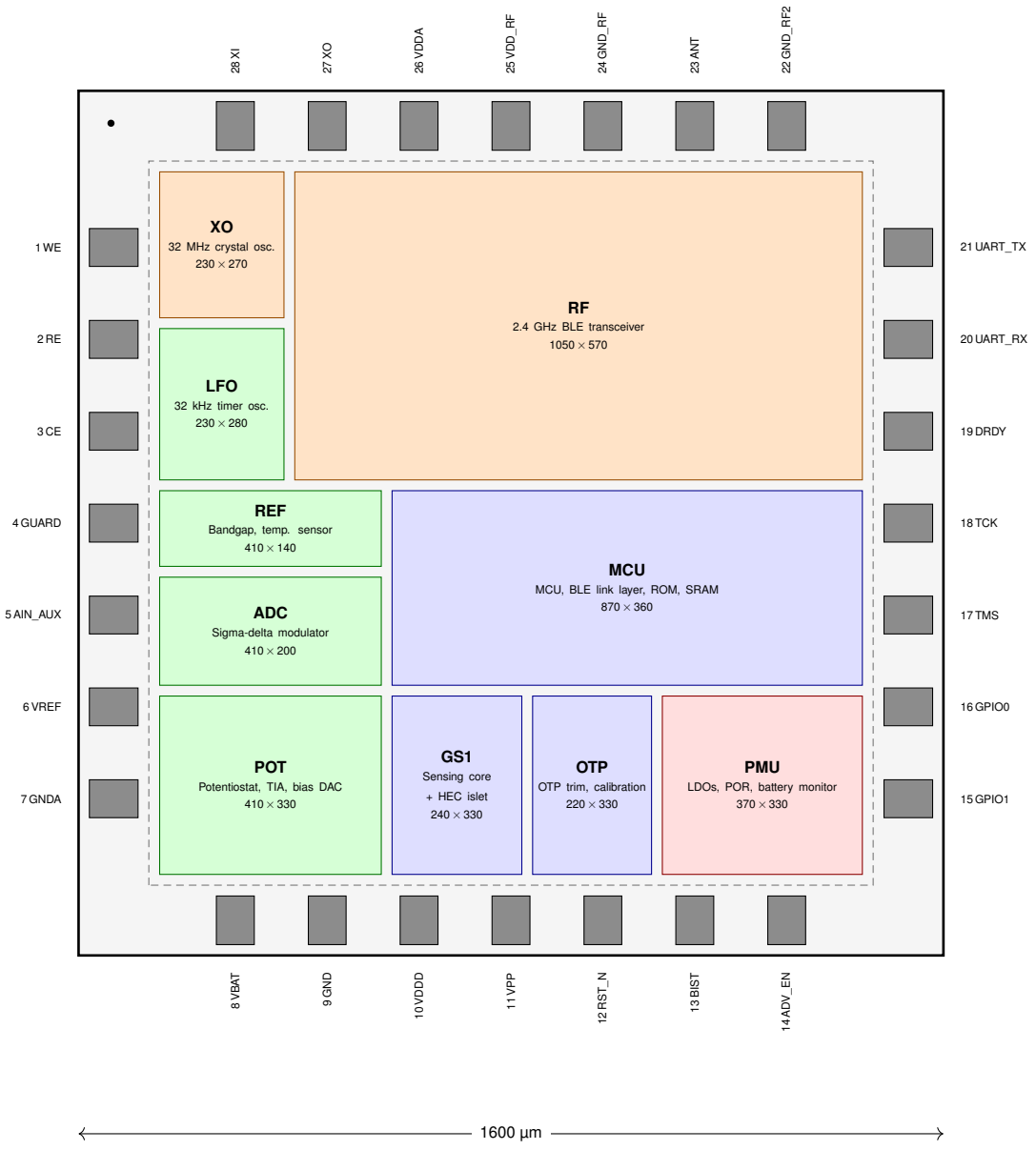


Figure 5: GS-1 floorplan (dimensions in µm; dot marks pin 1). Green: analog. Orange: RF. Blue: digital. Red: power. GDSII abstract and coordinate tables: `design/floorplan` in the repository.

Placement rules followed: sensor inputs on the left edge next to the potentiostat and far from the radio; antenna pad on the top edge between two RF grounds; digital pads on the right edge; the analog blocks in their own isolated well with a guard ring; the crystal oscillator next to its pads.

Pin list.

Table 2: Blocks: position (lower-left and upper-right corner, µm, origin at the lower-left die corner), area and status. Areas of analog and RF blocks are budgets.

Block	Function	Lower left	Upper right	mm^2	Status
POT	Potentiostat, transimpedance amplifier, bias DAC	150, 150	560, 480	0.135	specified
ADC	Second-order sigma-delta modulator, input multiplexer	150, 500	560, 700	0.082	specified
REF	Bandgap reference, temperature sensor	150, 720	560, 860	0.057	specified
LFO	32 kHz timer oscillator	150, 880	380, 1160	0.064	specified
XO	32 MHz crystal oscillator	150, 1180	380, 1450	0.062	specified
RF	2.4 GHz Bluetooth LE transceiver	400, 880	1450, 1450	0.599	specified
MCU	Processor, link layer, ROM, SRAM	580, 500	1450, 860	0.313	architecture
GS1	Sensing core and HEC islet	580, 150	820, 480	0.079	**RTL verified, synthesised**
OTP	Trim and calibration memory	840, 150	1060, 480	0.073	specified
PMU	Regulators, power-on reset, battery monitor	1080, 150	1450, 480	0.122	specified
	Total of blocks			1.59	

Pin	Name	Side	Function
1	WE	left	Working electrode (sensor current input)
2	RE	left	Reference electrode
3	CE	left	Counter electrode
4	GUARD	left	Guard ring driver for the WE track
5	AIN_AUX	left	Spare analog input (external NTC)
6	VREF	left	Reference bypass capacitor
7	GNDA	left	Analog ground
8	VBAT	bottom	Battery, 1.8 to 3.6 V
9	GND	bottom	Ground
10	VDDD	bottom	Digital core supply (decoupling)
11	VPP	bottom	OTP programming supply
12	RST_N	bottom	Reset, active low
13	BIST	bottom	Self-test enable
14	ADV_EN	bottom	Advertising enable
15	GPIO1	right	General purpose
16	GPIO0	right	General purpose
17	TMS	right	Debug data
18	TCK	right	Debug clock
19	DRDY	right	Data-ready / wake output
20	UART_RX	right	Commands / debug
21	UART_TX	right	Measurement frames / debug
22	GND_RF2	top	RF ground
23	ANT	top	2.4 GHz antenna, 50 ohm
24	GND_RF	top	RF ground
25	VDD_RF	top	RF supply (LDO output, decoupling)
26	VDDA	top	Analog supply (LDO output, decoupling)
27	XO	top	32 MHz crystal
28	XI	top	32 MHz crystal

Electrical targets.

Parameter	Target	Basis
Sensor bias (WE to RE)	0 to 1.0 V, 6-bit, default 0.6 V	register, verified
Current range	$\pm$200 nA full scale	digital scaling, verified
Current quantisation	0.76 pA	filter, verified
Current error, digital chain	$\leq$ 12.5 pA over $\pm$180 nA; $\leq$ 2.7 pA within $\pm$30 nA	model, ideal modulator
Input-referred noise	$<$ 10 pA rms	analog design target
Conversion time	4.1 ms per channel; 16 ms per reading in total	verified
Temperature	0.01 °C steps, on-chip sensor	target
Radio	Bluetooth LE 1 Mbit/s GFSK, 0 dBm, channels 37–39 in beacon mode	baseband verified
Average current, 1 reading/min	about 5 µA	estimate
Average current, 1 reading/s	about 14 µA	estimate

The current estimates assume 3 µA for the always-on sensor bias, 0.5 µA for timer and leakage, and 5 mA for 1.3 ms of

radio activity per reading; they are to be replaced by simulation and measurement. At 5 µA a 75 mAh coin cell outlasts a 14-day sensor many times over, so the cell can be made much smaller.

What the foundry receives. A foundry manufactures from a GDSII file drawn with its own process design kit and checked against its rules. For this technology the kit is released only to an institution under a non-disclosure agreement.

Item	State
Floorplan, die size, pad frame, pin list (GDSII abstract and tables)	**Done**
Digital core: Verilog, test benches, three matching models, gate-level netlist (SkyWater 130 nm)	**Done**
Digital core: sign-off layout in SkyWater 130 nm by the open flow (one automated run)	package ready
Analog and RF block specifications and area budgets	**Done** (this document)
Transistor-level design and layout of analog and RF blocks in the foundry kit	**Funded work**
Full-chip layout, design-rule and layout-versus-schematic checks, tape-out GDSII	**Funded work**

6 Methodology and work plan

6.1 Work packages

WP	Months	Tasks and deliverables (D)
WP1 Management, ethics, data	1–30	Coordination, reporting, data management plan, risk reviews. **D1.1** data management plan (M3); **D1.2** mid-term report (M15); **D1.3** final report (M30).
WP2 Digital core silicon and demonstrator	1–12	Layout run and submission on a SkyWater 130 nm shared wafer; FPGA prototype; wearable-size board with commercial front end and Bluetooth module; laboratory tests with glucose solutions. **D2.1** layout and sign-off reports (M2); **D2.2** FPGA demonstrator connected to the application (M3); **D2.3** silicon demonstrator and test report (M12).
WP3 Analog front end	3–18	Access to the 65 nm kit and tools; transistor-level potentiostat, sigma-delta modulator, references, regulators; 1 mm^2 test chip; measurement. **D3.1** design review (M9); **D3.2** tape-out (M10); **D3.3** measured noise, linearity and power (M18).
WP4 Radio and integration	10–30	Radio make-or-license decision; transceiver and processor block; full-chip integration and checks; tape-out; patch prototype. **D4.1** radio decision report (M12); **D4.2** GS-1 tape-out (M22); **D4.3** measured single-chip prototype (M30).
WP5 Islet model validation and software	1–30	Fit and test the glucose coupling of the HEC model on published data; in-vitro accuracy study; application releases. **D5.1** validation protocol with error bound (M6); **D5.2** validated model and paper (M18); **D5.3** in-vitro accuracy report and final application (M30).
WP6 Dissemination and exploitation	6–30	Publications, open releases, workshop, exploitation plan. **D6.1** open release of digital core and models (M12); **D6.2** exploitation and IP plan (M24); **D6.3** complete open release (M30).

6.2 Schedule and milestones

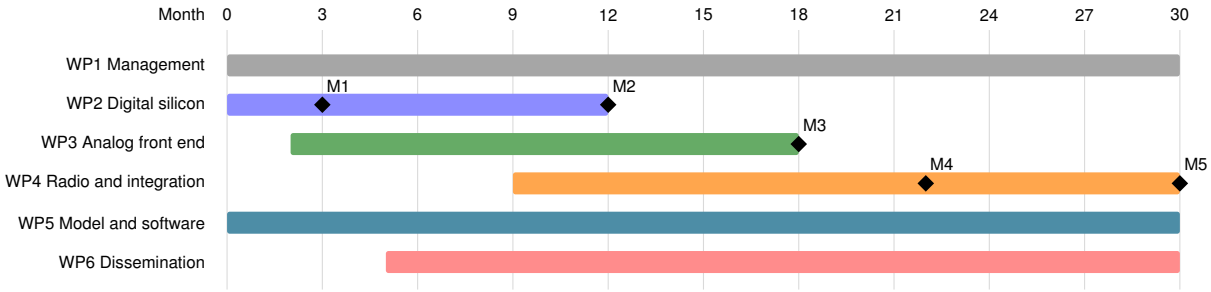


Figure 6: Schedule. Shared-wafer runs take four to nine months from submission to delivered chips, which sets the spacing of the milestones.

Milestone	Month	Verification
M1	3	FPGA prototype streams readings to the application over Bluetooth
M2 (go/no-go)	12	Fabricated digital core matches the reference model and works with the application
M3 (go/no-go)	18	Analog test chip meets the noise and accuracy criteria of O2
M4	22	GS-1 full-chip layout passes all sign-off checks; tape-out
M5	30	Single-chip prototype meets O3 and O5

WP4 integration spending is released only after M2 and M3. If M3 is missed, one corrective test-chip run is planned from contingency before integration.

6.3 Methods

Design. Digital blocks are written in Verilog and verified against executable reference models before layout, as already done for the core. Analog blocks are designed at transistor level with corner and Monte-Carlo simulation, laid out by hand and verified by post-layout simulation. **Measurement.** Noise and linearity are measured with a calibrated current source and a laboratory potentiostat as reference; glucose accuracy is measured in buffered glucose solutions against a laboratory analyser, using the ISO 15197 limits [11] as the criterion. **Model validation.** The islet model's coupling is fitted on one part of the published meal-test data and tested on the held-out part, with the error bound fixed in advance (D5.1).

7 Impact

Scientific. An open sensor-interface chip with a real-time physiological model; a reusable platform for sensor-chemistry and algorithm research; two journal papers and an open dataset. **Societal.** A route to earlier detection of impaired glucose regulation, relevant to the 252 million people with undiagnosed diabetes [1]; contributes to United Nations Sustainable Development Goal 3 (good health and well-being). **Capacity building.** Two engineers trained through a complete mixed-signal chip cycle from specification to measured silicon; course material from the open design. **Economic.** A documented chip and protocol lower the entry cost for local sensor and device developers; the islet-model technique can be licensed for other wearables.

7.1 Dissemination, exploitation and intellectual property

- *Publications:* two open-access journal papers (chip and model validation) and one conference presentation.
- *Open releases:* digital core, models and application under permissive open licences at month 12; complete design data that the foundry agreement allows at month 30.
- *Intellectual property:* the HEC model is published prior work of the applicant. Before each open release the host institution decides whether the on-chip islet technique is to be protected first; the release dates above leave time for that decision.
- *Users:* a workshop for sensor and clinical research groups in year 3; the application remains freely usable in the browser.

8 Data management and open science

Design sources, test benches, models, measurement scripts and raw measurement data are kept under version control in the project repository and released following the FAIR principles: findable through the repository and a DOI archive, accessible without registration, in open formats (Verilog, CSV, GDSII where the foundry agreement permits), and reusable under stated licences. No personal data are collected in this project.

9 Ethics, safety and regulatory position

All experiments are laboratory experiments on electronic circuits and glucose solutions. No human participants, human samples or animals are involved, so no ethical approval is required for the work proposed. Any later study on volunteers is outside this proposal and would need approval from the competent ethics committee. GlucoSense is a research platform and is labelled as not being a medical device; no clinical claim is made. The route to a regulated product (for example ISO 13485 quality management and medical-device approval) is described in the exploitation plan, D6.2.

10 Risk management

Risk	Likelihood	Impact	Mitigation
Access to the 65 nm kit and shared-wafer service needs an eligible institution	Medium	High	Apply through the host in month 1; alternatives are other shared-wafer brokers or a 180 nm process (larger radio)
Radio design slips or underperforms	Medium	High	Sensor chip remains useful with a commercial Bluetooth module (WP2, WP3); licensed radio as fallback, decided at M12
Analog noise target missed on first silicon	Medium	Medium	Dedicated test chip before integration; one corrective run from contingency
Shared-wafer schedule slips	Medium	Medium	FPGA and discrete-part prototypes keep WP5 and the application moving
Digital core fails the open flow's fixed 20 ns timing check	Low	Low	The design needs 125 ns; the long path is already split; further pipelining is routine
Glucose coupling of the islet model does not match physiology	Medium	Medium	Thresholds and gains are registers; validation with an error bound fixed in advance; the glucose measurement itself does not depend on it
Loss of key staff	Low	High	Complete, version-controlled documentation; executable reference models for every block

11 Resources and budget

11.1 Team

Role	Responsibility	Person-months
Principal investigator (Dr A. Mehmood)	Direction; HEC model and its validation; digital architecture; application	9
Analog/RF integrated-circuit engineer (to be recruited)	Front end, references, power management, radio	27
Digital and embedded engineer (to be recruited)	Digital back end, processor block, firmware, test boards	24

The principal investigator developed the HEC model [4, 5, 6, 7], the GlucoSense application and the verified digital core, and has taken two chips to clean layouts with the open flow [16]. Host institution to provide: laboratory space, electronic test equipment, and institutional access to the design-tool and shared-wafer programmes.

11.2 Fabrication and prototyping budget

Item	EUR	Source
WP2: digital core on a shared wafer, 8 tiles, with test board	900	estimate [15]
WP2: FPGA board, potentiostat front-end board, Bluetooth modules, boards, electrodes, solutions	3,500	estimate
WP3: TSMC 65 nm LP shared-wafer run, 1 mm^2	4,491	price list [14]
WP3: packaging and test boards	2,500	estimate
WP4: TSMC 65 nm LP shared-wafer run, 2.56 mm^2 (4,491 + 15.6 $\times$ 419)	11,027	price list [14]
WP4: packaging, patch boards, antennas	4,500	estimate
Design-tool and kit access for 30 months (academic programme)	7,500	estimate
WP5: reference instruments, laboratory potentiostat access, analyser consumables	3,000	estimate
WP6: open-access publication fees, one conference, workshop	3,500	estimate
Subtotal	40,918	
Contingency 12 % (covers one corrective test-chip run)	4,910	
Total, fabrication and prototyping	**45,828**	

Justification. Silicon is bought in the smallest available units of shared-wafer services, which is the lowest-cost route to real chips; the two-step approach (test chip, then integration) spends 4,491 EUR to remove the main technical risk before the 11,027 EUR run. Price-list figures are the standard Europractice 2026 prices read on 9 October 2026 and must be confirmed by quotation; estimates are the applicant's own. **Staff costs are not included** and are to be added at the host institution's scales for the person-months above. If the radio is licensed instead of designed, a licence fee replaces part of the engineer time and must be quoted.

12 Status declaration: completed and funded work

Completed and available in the repository (https://github.com/asifnazquaidian/glucosense-gs1): Verilog of the sensing core and islet block (src, design/rtl); test benches and reference models (test, design/tb, design/model); comparison reports, synthesis report and netlist (design/results); floorplan GDSII abstract with pad and block tables (design/floorplan); the application (app), also running at https://glucosensor.netlify.app.

Not yet done: no chip has been fabricated; the digital-core layout run has not been completed; analog, radio, processor and memory blocks are not designed; the firmware of the first ESP32 prototype has not been run on hardware; Bluetooth reception has not been tested over the air; nothing has been tested with a real sensor or on people. Where applicable, AI-assisted tools were used for convenience in preparing design resources. Independent review by the funded engineers is part of WP2.

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